

The numbers 'e' and 'n' make up the public key, while 'd' and 'n' make up the private key. The calculation is given as follows ('m' is the message to be encrypted and 'c' is the ciphertext produced after encryption):

For encryption:

$$c = m^e \pmod{n}$$

For decryption:

$$m = c^d \pmod{n}$$

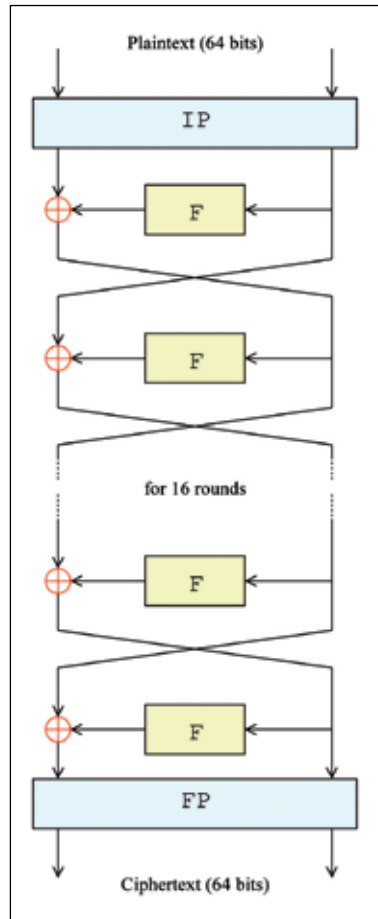
Note: the character ^ refers to the power operator as in $2^2 = 4$ and $2^3 = 8$

Now, this is applicable for numbers and messages are not always numbers. However, with digital messages it's different because any and everything is represented in binary form which can be directly translated to a number.

The strength of the cipher depends on how difficult it's to guess the numbers p and q. Once p and q are known, it would not take too much effort to find the rest of the numbers. Since the product of p and q is n and n is a part of public key which is used for encryption and can be well known by third

parties, the only way to know p and q is to find out the factors of n. Interestingly enough, n can have only 4 factors: 1, n, p and q. The first two are applicable for any number and thus one has to find p and q. The strength of RSA, as such, depends on p and q. Hence they both should be very large prime numbers. This makes the factorization of n extremely difficult (since there are no easy ways known to factorize the products of large prime numbers).

Algorithms are what protect the secrecy of the message and the strength of the cipher depends on two things: the key used and the algorithm used.



Overall DES Algorithm